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Natural Language Processing (S1-24_AIMLCZG530) WEBINAR 3

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BERT and Introduction to LLM

Agenda



- Overview of BERT
- Key Concepts
- Architecture
- Pre-Training
- Fine Tuning and Applications
- Introduction to LLM
- References



Overview of BERT

- BERT stands for **B**idirectional **E**ncoder **R**epresentations from **T**ransformers.
- BERT uses encoder only Transformer architecture
- It is a method of pre-training language representations and is pre-trained on a large corpus of unlabelled text including the entire Wikipedia(that's 2,500 million words!) and Book Corpus (800 million words)
- BERT had been adopted by Google Search for over 70 languages since 2019
- BERT is bidirectional i.e learning happens in left to right and right to left

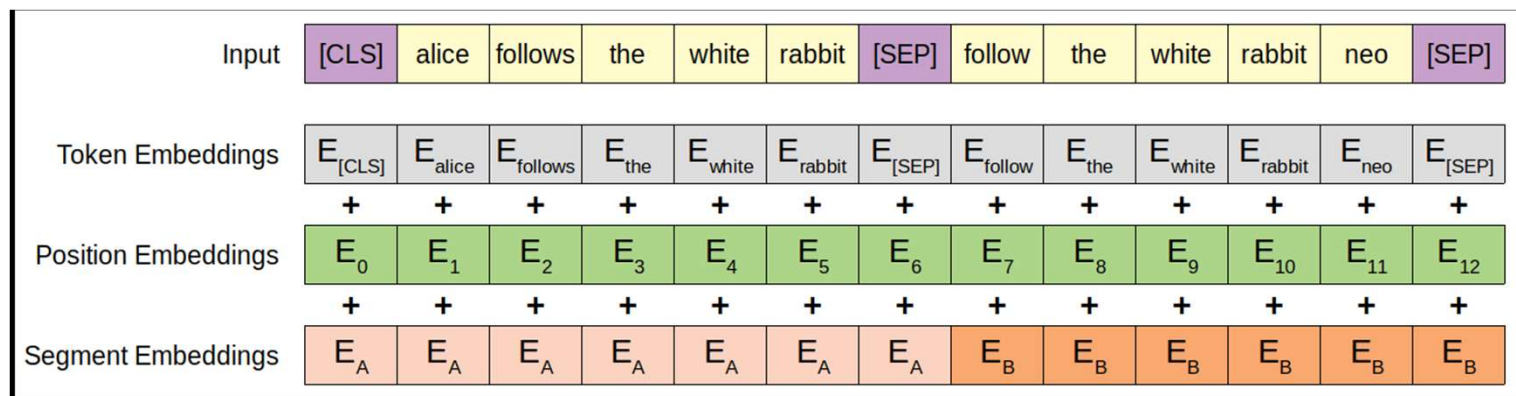


Key Concepts

- BERT is pre-trained on two NLP tasks:
 - Masked Language Modeling
 - Next Sentence Prediction
- Context-free models such as [word2vec](#) or [GloVe](#) generate a single "word embedding". E.x 'bank' has the same vector embedding in context of computer bank deposits and river bank
- BERT effectively addresses contextual embeddings because its bidirectional.
- E.x 'I made a bank deposit', here embedding for bank is learned in the context of 'I made a' and 'deposit'

Architecture

- BERT is a Transformer based model.
- 3 kinds of embeddings are used in BERT, Token Embeddings, Position Embeddings, Segment Embeddings.



Special Tokens in BERT:

[CLS] : The start token of every document.

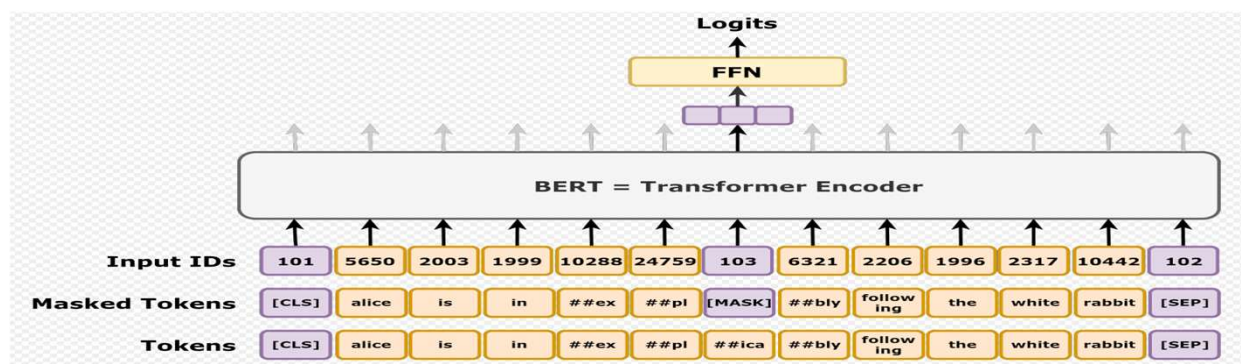
[SEP] : Placed between each sentence.

[PAD] : Added at the end of the document to keep the length of document to 512

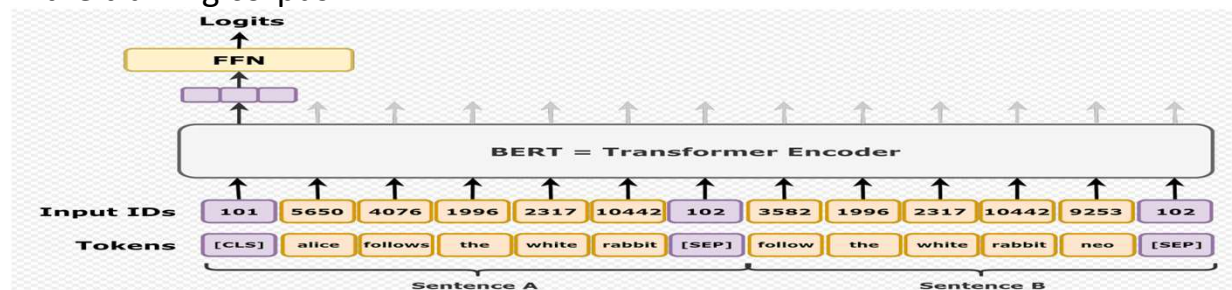
##token : Indicates the start of a "word piece."

Pre-Training

- BERT is trained on English Wikipedia and Books corpus.
- BERT is pretrained on two tasks, Masked language Modeling and next sentence Prediction
- In Masked Language Modeling, 15% of tokens would be randomly selected for masked-prediction task, and the training objective was to predict the masked token given its context.



- In Next Sentence Prediction, given a span of two texts, model predicts if the two appears sequentially in the training corpus.





Fine-Tuning and Applications

- Pre-Trained BERT can be used for various NLP applications
- Pre-Trained BERT can be used as embedding layer in Deep Neural networks
- Most common use cases for BERT are, Sentiment Analysis, Question Answering, Text Classification, POS Tagging, Sentence Classification.
- Since BERT is a bidirectional Transformer model, trained on Masked Language Model, it is not suited for Generative tasks like Next word prediction like GPT
- GPT (Generative Pre-trained Transformers) are unidirectional and auto-regressive model which are ideal for next word prediction and text generative applications.



Introduction to LLM

- Large Language Models, pre-trained on large amounts of data.
- Trained with self-supervised learning approach.
- Some of the recent LLM's are:
 - GPT-4 -
 - DeepSeek R1 – 671 billion parameters
 - Llama 3.1 – 405 billion parameters
- Some of the applications of LLM's :
 - Code generation
 - Information extraction
 - Question Answering
 - Content Creation



References

- BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding [<https://arxiv.org/abs/1810.04805>]
- <https://github.com/google-research/bert>
- [https://en.wikipedia.org/wiki/BERT_\(language_model\)](https://en.wikipedia.org/wiki/BERT_(language_model))



Code Walkthrough



Thank you